ALPHA SYSTEM REPLACEMENT

GDPS Production Cutover Operations Runbook

Legacy Alpha SMB Namespace → GDPS Production File Cluster

Project	PR-26-GPO-2540 — Alpha System Replacement
System	GDPS
Document	Production Cutover Operations Runbook
Classification	UNCLASSIFIED
Governing Standard	GDPS Windows File Cluster and Legacy SMB Namespace Standard
Hardware Baseline	GDPS Hardware Infrastructure Master Design
Status	Execution Runbook v1.2 - Cross-Data-Center Failover

Purpose: provide an executable, evidence-driven production switchover procedure that Operations can follow without improvisation.

Executive Summary

This runbook converts the approved GDPS cluster, SMB namespace, storage, replication, and cutover standards into an operational sequence. It is written for the production change window and is intended to be followed step by step by Infrastructure, Storage, Network/AD/DNS, GDPS application staff, and legacy application owners.

The production cutover is controlled by two hard facts. First, existing applications cannot be changed and must continue to consume the existing UNC interfaces under \\alpha0, \\alpha3, and \\alpha6. Second, the legacy Alpha environment and GDPS must never advertise the same production SMB identity at the same time. Therefore, all GDPS pre-cutover work is performed through temporary identities such as alpha0-1, alpha3-1, and alpha6-1.

The actual cutover occurs only after production writes are frozen, final synchronization is validated, and the old Alpha production identities have been proven inactive. Only then may the real alpha0, alpha3, and alpha6 names be activated as clustered SMB client-access identities on the GDPS production file cluster.

TEST is the mandatory rehearsal environment. The complete cutover process must be exercised in TEST using TEST-specific or temporary identities before the production change. The sandbox is not required to establish production readiness because DEV and TEST collectively exercise the full GDPS operating model. Because HQ and IM are separate data centers, TEST rehearsal must also prove that the clustered file/service capability can continue at the surviving site after complete loss of either HQ or IM.

HARD STOP RULE: alpha0, alpha3, and alpha6 must not be configured, registered, advertised, resolvable to GDPS, or usable from GDPS before the authorized production cutover. The change lead must stop the cutover if duplicate ownership is detected.

Control	Value	Operational Meaning
Pre-cutover names	alpha0-1 / alpha3-1 / alpha6-1	GDPS staging, replication, validation, rehearsal
Production names	alpha0 / alpha3 / alpha6	Legacy Alpha only until identity-transfer gate; GDPS only after gate passes
PROD file cluster	HQVMGDPSFSP1 / IMVMGDPSFSP2	Cross-data-center Windows Server Failover Cluster: HQVMGDPSFSP1 at HQ + IMVMGDPSFSP2 at IM
Quorum	File Share Witness	Independent of clustered data storage
Production storage	14.4 TB usable minimum	Shared enterprise storage, redundant paths / MPIO
COOP	Replicated recovery target	Two-node file cluster in crate; does not advertise production names during normal PROD operation

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1. Runbook Authority and Scope

This runbook is the operational execution document for switching the production Alpha SMB/file-service interfaces to the GDPS production file cluster. It does not authorize business-rule changes, share-name redesign, path redesign, or application changes. If the runbook encounters a condition that requires one of those changes, the cutover stops.

- Follow the steps in order. Do not skip a gate because a later step appears to work.
- Record evidence as each step completes.
- Only the designated Cutover Lead may authorize progression through a GO/NO-GO gate.
- Any owner may call a STOP when a defined stop condition occurs.
- Commands shown in this runbook are validation/administration patterns. Site-specific object names, witness paths, DNS records, service accounts, and AD/SPN values must be filled in before execution.

Cross-data-center precondition: PROD spans HQVMGDPSFSP1 at the HQ data center and IMVMGDPSFSP2 at the IM data center. Production cutover shall not proceed until the equivalent TEST pair has demonstrated node failover and full-site failover in both directions (HQ lost -> IM survives, and IM lost -> HQ survives), including witness/quorum, storage/data access, SMB service availability, and application validation.

2. Roles, Command Authority, and Communications

Role	Primary Responsibility	Authority
Cutover Lead	Controls sequence, calls GO/NO-GO, owns change timeline and rollback decision	Proceed / Stop / Roll back
Windows/Cluster Administrator	WSFC, clustered SMB roles, witness, cluster failover, SMB service identity activation	Execute cluster/file-service changes
Storage Engineer	Replication, shared storage, MPIO/fabric, final sync and storage health	Execute/validate storage operations
Network/DNS/AD Administrator	DNS/name ownership, required AD/Kerberos/SPN work, cache/name validation	Execute identity/network changes
GDPS Application Lead	File Processor, Middleware, Delivery Worker, service validation	Start/stop/validate GDPS runtimes
Alpha Operations Lead	Stop Alpha processing/writes; validate old identities are no longer serving	Control legacy source
Legacy Application Testers	Run representative unchanged applications against production UNC paths	Consumer acceptance
Monitoring/Operations	Observe cluster, storage, SMB, database, application, replication health	Stabilization monitoring

CHANGE BRIDGE: All named owners must be present or represented on the cutover bridge before Phase B begins. No production identity transfer occurs by email-only coordination.

3. Environment and Identity Reference

Object	Name	Rule
GDPS PROD cluster node 1	HQVMGDPSFSP1	Physical node; never renamed to Alpha identity

GDPS PROD cluster node 2	IMVMGDPSFSP2	Physical node; never renamed to Alpha identity
Temporary migration identity	alpha0-1	Pre-cutover only
Temporary migration identity	alpha3-1	Pre-cutover only
Temporary migration identity	alpha6-1	Pre-cutover only
Legacy production identity	alpha0	Alpha before cutover; GDPS only after identity-transfer gate
Legacy production identity	alpha3	Alpha before cutover; GDPS only after identity-transfer gate
Legacy production identity	alpha6	Alpha before cutover; GDPS only after identity-transfer gate

3.1 Required Pre-Filled Site Values

Required Value	Production Value
PROD cluster name / administrative identity	_____
PROD File Share Witness UNC	_____
PROD shared-storage replication job/policy	_____
Legacy alpha0 IP/DNS details	_____
Legacy alpha3 IP/DNS details	_____
Legacy alpha6 IP/DNS details	_____
GDPS alpha0 target IP/client-access details	_____
GDPS alpha3 target IP/client-access details	_____
GDPS alpha6 target IP/client-access details	_____
Authoritative DNS servers	_____
AD domain / service identity	_____
Monitoring dashboard / event location	_____

4. Production Cutover Gates

Gate	Decision	Required Evidence	Authority
G0	TEST rehearsal complete	TEST cutover rehearsal completed, including HQ-loss and IM-loss site-failover scenarios; defects closed or formally accepted.	Cutover Lead
G1	PROD infrastructure ready	WSFC healthy; witness online; shared storage/MPIO healthy; replication healthy	Cluster + Storage
G2	Production freeze achieved	Alpha processing/writes stopped; open sessions/locks resolved	Alpha Ops
G3	Final synchronization validated	Final data sync complete; integrity evidence accepted	Storage + GDPS
G4	Legacy identities released	alpha0/alpha3/alpha6 no longer served by legacy Alpha; no duplicate name ownership	Alpha + Network
G5	GDPS identities activated	alpha0/alpha3/alpha6 served only by GDPS clustered file service	Cluster + Network
G6	Consumer validation passed	Legacy apps work unchanged; GDPS runtimes healthy	GDPS + App Testers
G7	Production released	Processing resumed; monitoring stable; rollback window controlled	Cutover Lead

5. Phase A — Pre-Cutover Readiness

Step	Owner	Action	Command / Check	Expected Result	Evidence	STOP / Rollback Trigger
A01	Cutover Lead	Open change bridge; verify attendance and authority.	Call roll; identify named owners and backups.	All required roles present.	Bridge roster.	Missing owner with required execution authority.
A02	Cluster Admin	Verify PROD cluster service and both site-separated nodes are healthy.	Get-Cluster; Get-ClusterNode; Get-ClusterGroup; verify HQVMGDPSFSP1 at HQ and IMVMGDPSFSP2 at IM.	Cluster online; both nodes Up at their intended data centers; no unexpected failed roles.	Command output / screenshot.	Any node/role unexpectedly Down or Failed.
A03	Cluster Admin	Verify File Share Witness quorum.	Get-ClusterQuorum	Quorum type and configured File Share Witness match runbook; witness remains independent of clustered data and is reachable from the surviving site if HQ or IM is lost.	Command output.	Witness offline, inaccessible, or on clustered data storage.
A04	Storage	Verify all shared-storage paths and multipath state.	Use approved MPIO/storage health tools; confirm both fabrics/paths.	No degraded path; shared storage accessible from both nodes.	Storage health evidence.	Any required path/fabric unavailable.
A05	Storage	Verify replication is healthy and current.	Use storage replication console/CLI defined in site values.	Replication running; lag within accepted pre-cutover threshold.	Replication status.	Replication broken or lag outside threshold.
A06	Cluster Admin	Verify temporary identities are online and final production identities are absent from GDPS.	Get-SmbShare; Resolve-DnsName alpha0-1/3-1/6-1; verify alpha0/3/6 do not point to GDPS.	Temporary names work; production names do not point to GDPS.	Command outputs.	Any final production name already resolves/answers from GDPS.
A07	GDPS App	Validate required share tree and permissions through temporary names.	Test-Path \\alpha0-1\<share>; repeat for inventory; use approved ACL validation.	All required shares and subpaths available.	Checklist / ACL evidence.	Missing share/path or incorrect ACL.
A08	Legacy App Testers	Run representative application checks using temporary/test identities where feasible.	Application-specific test cases.	Required read/write behavior succeeds.	Test results.	Critical application behavior fails.
A09	GDPS App	Verify File Processor, Middleware, Delivery Worker, and PostgreSQL health.	Approved service/health endpoints and service status checks.	All GDPS runtime components healthy.	Health evidence.	Any production-critical GDPS service unhealthy.

A10	Cutover Lead	Review TEST rehearsal results and open defects.	Review rehearsal evidence and defect list.	G0 satisfied.	Signed GO record.	Critical unresolved rehearsal defect.

6. Phase B — Production Freeze

Step	Owner	Action	Command / Check	Expected Result	Evidence	STOP / Rollback Trigger
B01	Cutover Lead	Announce production freeze start and timestamp.	Record T0 in bridge log.	All teams acknowledge freeze.	Timestamped bridge log.	Any team continues unauthorized production writes.
B02	Alpha Ops	Stop/suspend Alpha processing that creates or modifies files in cutover scope.	Use approved Alpha/POLLVAX operational stop procedure.	No new Alpha processing writes occur.	Alpha process status.	Unable to stop a writer.
B03	Integration Owners	Stop upstream/downstream writers that must be quiesced for final sync.	Use system-specific stop/pause procedure.	All required external writers quiesced.	Owner confirmations.	Writer cannot be paused or continues changing files.
B04	Alpha Ops	Check for open sessions, locks, and incomplete processing work.	Use Alpha/SMB/file-session tools appropriate to source.	No unresolved write locks or in-flight jobs in cutover scope.	Session/job evidence.	Active write lock/job cannot be resolved.
B05	Cutover Lead	Declare production source frozen.	Confirm B02–B04 complete.	G2 satisfied.	GO decision record.	Any uncontrolled writer remains.

7. Phase C — Final Data Synchronization

Step	Owner	Action	Command / Check	Expected Result	Evidence	STOP / Rollback Trigger
C01	Storage	Capture final Alpha source checkpoint.	Record replication snapshot/checkpoint, source time, or agreed marker.	Final source checkpoint established.	Checkpoint evidence.	Unable to establish authoritative source point.
C02	Storage	Run final synchronization to GDPS production storage.	Execute approved replication/sync job.	Final sync completes successfully.	Replication job result.	Job failure or unresolved error.
C03	Storage	Verify replication lag is zero/accepted final state.	Replication platform status.	Target is current to final checkpoint.	Status output.	Target not current.
C04	GDPS + Storage	Validate file counts and provisioned scope.	Compare required source/target inventories by share/volume.	Counts reconcile within defined rules.	Comparison report.	Unexpected missing/extra critical files.
C05	GDPS + Storage	Validate sizes/timestamps/checksums per agreed integrity method.	Run approved comparison/hash checks on defined sample/full scope.	Integrity evidence accepted.	Validation report.	Mismatch requiring source/target reconciliation.
C06	Cutover Lead	Approve final synchronized dataset.	Review C01–C05 evidence.	G3 satisfied.	GO record.	Any unresolved data-integrity issue.

8. Phase D — Legacy Alpha Identity Shutdown

NO OVERLAP: Do not activate alpha0, alpha3, or alpha6 on GDPS until every applicable legacy Alpha identity has been disabled and independently verified as no longer serving SMB/name traffic.

Step	Owner	Action	Command / Check	Expected Result	Evidence	STOP / Rollback Trigger
D01	Alpha Ops	Disable legacy SMB/service exposure for alpha0.	Approved legacy shutdown/network procedure.	alpha0 no longer serves legacy SMB.	Source status.	alpha0 still answers from legacy source.
D02	Alpha Ops	Disable legacy SMB/service exposure for alpha3.	Approved legacy shutdown/network procedure.	alpha3 no longer serves legacy SMB.	Source status.	alpha3 still answers from legacy source.
D03	Alpha Ops	Disable legacy SMB/service exposure for alpha6.	Approved legacy shutdown/network procedure.	alpha6 no longer serves legacy SMB.	Source status.	alpha6 still answers from legacy source.
D04	Network/DNS	Remove/disable old DNS/name registrations as required by the approved identity design.	DNS/AD administrative procedure defined for site.	Old registrations no longer direct clients to Alpha.	DNS change evidence.	Stale authoritative record remains.
D05	Network/DNS	Verify no duplicate production identity exists.	Resolve-DnsName alpha0/alpha3/alpha6 from multiple representative clients; test legacy IPs if required.	No production identity resolves to/answers from old Alpha.	Command outputs.	Any duplicate/stale identity remains.
D06	Cutover Lead	Authorize GDPS production identity activation.	Review D01–D05 evidence.	G4 satisfied.	GO record.	Any old identity still active.

9. Phase E — GDPS Production Identity Activation

Step	Owner	Action	Command / Check	Expected Result	Evidence	STOP / Rollback Trigger
E01	Cluster Admin	Activate GDPS clustered SMB client-access identity alpha0 using the prebuilt cutover configuration.	Execute reviewed WSFC/SMB client-access activation procedure.	alpha0 role/name online on GDPS cluster.	Cluster role output.	Activation fails or identity binds to wrong node/service.
E02	Cluster Admin	Activate alpha3 clustered SMB client-access identity.	Reviewed WSFC/SMB activation procedure.	alpha3 online on GDPS cluster.	Cluster role output.	Activation fails.

E03	Cluster Admin	Activate alpha6 clustered SMB client-access identity.	Reviewed WSFC/SMB activation procedure.	alpha6 online on GDPS cluster.	Cluster role output.	Activation fails.
E04	Network/DNS/AD	Create/enable final production name registrations and required authentication bindings.	Execute reviewed DNS/AD/Kerberos procedure.	Names resolve only to GDPS service; authentication succeeds.	DNS/AD evidence.	Duplicate, wrong target, or auth failure.
E05	Cluster Admin	Verify shares are online beneath each final identity.	Get-SmbShare; Test-Path \\alpha0\<share>, etc.	All required shares accessible.	Share inventory output.	Required share missing/offline.
E06	Network/DNS	Verify name resolution from multiple production subnets/representative clients.	Resolve-DnsName alpha0; alpha3; alpha6; clear/test caches as approved.	Consistent GDPS resolution.	Client-side outputs.	Inconsistent/stale resolution.
E07	Cutover Lead	Approve production identity transfer complete.	Review E01–E06.	G5 satisfied.	GO record.	Any name/share/auth issue unresolved.

10. Phase F — Application and Infrastructure Validation

Step	Owner	Action	Command / Check	Expected Result	Evidence	STOP / Rollback Trigger
F01	Legacy App Tester	Open/read representative legacy path under \ \alpha0.	Application test or Test-Path with representative account.	Read succeeds without application config change.	Test result.	Cannot read required path.
F02	Legacy App Tester	Open/read representative path under \\alpha3.	Application test.	Read succeeds.	Test result.	Failure.
F03	Legacy App Tester	Open/read representative path under \\alpha6.	Application test.	Read succeeds.	Test result.	Failure.
F04	Legacy App Tester	Perform authorized write/update test on required writable share.	Application-specific controlled write test.	Write succeeds; expected permissions enforced.	Test evidence.	Write denied/incorrect permission or unauthorized write allowed.
F05	GDPS App	Validate File Processor access to production shared locations.	Start/health check; controlled processing validation.	File Processor sees configured production paths and processes approved validation item.	Logs/status.	Path/access/processing failure.
F06	GDPS App	Validate Middleware/API, Issuer, Delivery Worker, PostgreSQL.	Approved health endpoints/service checks.	All core services healthy.	Health evidence.	Critical service unhealthy.
F07	Cluster Admin	Validate cluster role ownership and witness state.	Get-ClusterGroup; Get-ClusterQuorum; Get-ClusterNode.	Roles online; witness healthy; both nodes healthy.	Command output.	Unexpected failed resource or quorum warning.
F08	Storage	Validate shared storage and replication to COOP.	Storage/replication status tools.	All paths healthy; PROD→COOP replication active.	Status evidence.	Replication/path failure.
F09	Cluster Admin	If authorized in change plan, perform controlled file-service failover test.	Move-ClusterGroup <file-role> -Node <peer>; then validate SMB path.	Legacy UNC remains available through failover within accepted interval.	Failover evidence.	UNC unavailable beyond threshold or resource fails.
F10	Legacy App Testers	Run defined representative legacy application suite	Application test script/checklist.	All critical consumers pass.	Signed acceptance.	Critical legacy consumer fails.

		unchanged.				
F11	Cutover Lead	Approve production service release.	Review F01–F10.	G6 satisfied.	GO record.	Critical defect unresolved.

11. Phase G — Release and Stabilization

Step	Owner	Action	Command / Check	Expected Result	Evidence	STOP / Rollback Trigger
G01	GDPS App	Enable/resume GDPS production processing in approved order.	Approved service start/resume procedure.	Processing resumes normally.	Service/log evidence.	Unexpected processing errors.
G02	Integration Owners	Resume upstream/downstream integrations in approved order.	System-specific resume procedure.	External flows resume without backlog/error escalation.	Owner confirmations.	Critical interface fails.
G03	Monitoring	Begin heightened monitoring window.	Monitor WSFC, witness, storage, SMB, PostgreSQL, application, replication metrics/events.	No critical alerts; trends stable.	Monitoring snapshots.	Critical or sustained degradation.
G04	Storage	Confirm PROD→COOP replication remains current after production writes resume.	Replication status.	Replication healthy and within accepted lag.	Status output.	Lag/failure outside threshold.
G05	Cluster Admin	Review SMB sessions and cluster events for abnormal errors.	Get-SmbSession; cluster event review.	Expected sessions; no material cluster/SMB errors.	Output/event record.	Repeated access/session/cluster failures.
G06	Cutover Lead	Declare cutover complete after stabilization criteria met.	Review monitoring window and open issues.	G7 satisfied.	Completion record.	Unresolved severity-1/critical condition.
G07	Cluster Admin	Retire temporary alpha0-1/alpha3-1/alpha6-1 identities when stabilization plan permits.	Reviewed removal procedure.	Temporary migration names removed/disabled.	Change evidence.	Dependency still unexpectedly uses temporary name.

12. Rollback Decision Tree and Procedure

ROLLBACK BRANCH: The critical branch is whether GDPS has accepted authoritative production writes. A simple name switch-back is permitted only before new authoritative GDPS writes have begun.

Step	Owner	Action	Command / Check	Expected Result	Evidence	STOP / Rollback Trigger

R0	Cutover Lead	Determine whether GDPS has accepted authoritative production writes.	Review processing/write-enable timestamp and logs.	Branch A or B selected.	Decision record.	Unclear authoritative state — STOP and reconcile before any identity change.
R-A1	GDPS App	If NO new authoritative GDPS writes: stop GDPS processing/writes.	Stop services/writers.	GDPS quiesced.	Service status.	Unable to quiesce.
R-A2	Cluster Admin	Disable GDPS alpha0/alpha3/alpha6 clustered identities.	Reviewed disable procedure.	GDPS no longer serves production names.	Cluster/name evidence.	Identity still active.
R-A3	Network/DNS	Remove/disable GDPS production name registrations.	DNS/AD procedure.	Names no longer resolve to GDPS.	DNS evidence.	Stale GDPS record remains.
R-A4	Alpha Ops + Network	Restore Alpha production identities and validate paths.	Legacy restoration procedure.	Legacy names resolve/serve from Alpha only.	Validation evidence.	Duplicate or failed identity.
R-A5	Cutover Lead	Authorize Alpha processing resume after source consistency check.	Review source and application validation.	Alpha resumed.	Rollback completion record.	Source inconsistent.
R-B1	GDPS App	If GDPS HAS accepted authoritative writes: freeze GDPS writes immediately.	Stop writers/services.	No further divergence.	Timestamp/evidence.	Unable to freeze.
R-B2	Storage + App	Preserve Alpha and GDPS datasets; do not overwrite either side.	Snapshot/protection procedure.	Both states retained.	Snapshot/protection evidence.	Any destructive overwrite risk.
R-B3	Cutover Lead	Initiate controlled data reconciliation plan.	Compare writes since identity transfer; establish authoritative dataset.	Reconciled recovery plan approved.	Reconciliation record.	No authoritative dataset can be determined.
R-B4	All	Only after reconciliation, execute controlled namespace return or continue GDPS recovery.	Approved reconciliation outcome.	Single authoritative environment restored.	Closure evidence.	Any dual ownership.

13. TEST Rehearsal Procedure

The TEST rehearsal is mandatory and must be treated as a production dress rehearsal, not a functional smoke test. The TEST physical file/service pair spans the separate HQ and IM data centers and uses TEST-specific client-access names and data; it does not use alpha0, alpha3, or alpha6. The rehearsal must demonstrate both ordinary node failover and complete data-center failover in each direction.

Step	Owner	Action	Command / Check	Expected Result	Evidence	STOP / Rollback Trigger
T01	All	Execute Phases A through G in TEST using TEST-specific names.	Use this runbook with TEST values substituted.	Sequence completes.	Rehearsal log.	Any critical step cannot be reproduced.
T02	Cluster Admin	Force controlled loss/failover of one TEST file-cluster node.	Move/stop approved test role/node action.	Clustered SMB service remains available on the peer node in the	Failover evidence.	Role fails or witness/quorum behaves unexpectedly.

				alternate data center.		
T03	Cluster Admin	Exercise File Share Witness/quorum behavior with HQ unavailable and with IM unavailable.	Approved witness test procedure.	Surviving TEST site establishes quorum and owns the clustered service in each approved site-loss scenario.	Witness/quorum evidence.	Unexpected loss of cluster service.
T04	Storage	Exercise storage/data availability after simulated loss of HQ and simulated loss of IM.	Approved storage test.	Surviving site retains or recovers access to the required TEST dataset within the approved recovery target.	Storage evidence.	Application I/O failure beyond target.
T05	Storage	Exercise replication interruption and recovery.	Approved TEST replication procedure.	Replication resumes and reconciles.	Replication evidence.	Unable to recover cleanly.
T06	App Testers	Run legacy application test suite through TEST/temp namespace.	Application test suite.	Critical workflows pass before and after HQ-to-IM and IM-to-HQ site failover.	Signed results.	Critical workflow fails.
T07	Cutover Lead	Exercise rollback Branch A in TEST.	Use rollback steps.	Clean return achieved.	Rollback evidence.	Rollback fails.
T08	Cutover Lead	Document rehearsal durations for production scheduling.	Record actual start/end times by phase.	Production timeline refined.	Timing report.	Unknown/unbounded critical step duration.

14. COOP Recovery Identity Procedure

Step	Owner	Action	Command / Check	Expected Result	Evidence	STOP / Rollback Trigger
C-O1	Recovery Lead	Declare PROD unavailable and authorize COOP recovery.	Incident/COOP declaration process.	Recovery authority established.	Declaration record.	PROD still actively owns production names.
C-O2	Storage	Confirm COOP replicated dataset is at accepted recovery point.	Replication/recovery status.	Recovery dataset accepted.	Recovery evidence.	Dataset not recoverable/accepted.
C-O3	Network/DNS	Prove PROD alpha0/alpha3/alpha6 identities cannot advertise/serve.	Name/network validation.	No duplicate owner possible.	Validation evidence.	PROD identity still active.
C-O4	Cluster Admin	Activate production SMB identities on COOP clustered file service.	Reviewed COOP activation procedure.	alpha0/3/6 served by COOP only.	Cluster evidence.	Activation/auth failure.
C-O5	App Testers	Validate representative legacy consumers.	Application tests.	Consumers operate unchanged.	Acceptance evidence.	Critical consumer fails.
C-O6	Recovery Lead	Release COOP processing and monitor.	Recovery release sequence.	COOP assumes service.	Recovery record.	Critical instability.

15. Evidence Package and Cutover Closure

Evidence Item	Owner	Minimum Content
Bridge/change log	Cutover Lead	Timestamps, GO/NO-GO decisions, incidents, owner confirmations
Cluster validation	Cluster Admin	Node state, groups, quorum/witness, role ownership, failover evidence
Storage validation	Storage	Path/fabric health, replication state, final sync, integrity comparison
DNS/identity evidence	Network/DNS/AD	Before/after name resolution, identity ownership, authentication validation
SMB/share inventory	Cluster Admin	Share list, ACL validation, representative Test-Path/access results
Application validation	GDPS + Legacy testers	File Processor/API/Delivery health plus unchanged legacy application test results
Rollback readiness	Cutover Lead	Decision point and evidence branch selected/not required
Stabilization report	Operations	Monitoring results for agreed stabilization window
Final acceptance	Cutover Lead	Formal cutover completion and residual issues

Appendix A — Command and Validation Reference

The commands below are operational patterns for Windows/WSFC validation. They must be executed with the appropriate administrative context and with site-specific names filled in. They do not replace change-control authorization.

Purpose	Command / Pattern	Expected Use
Cluster health	Get-Cluster	Returns intended PROD cluster; no unexpected connectivity error
Cluster nodes	Get-ClusterNode	Both expected nodes listed; state Up before cutover
Cluster groups	Get-ClusterGroup	Expected clustered file/service roles Online
Quorum/witness	Get-ClusterQuorum	File Share Witness shown and healthy
SMB shares	Get-SmbShare	Required GDPS/legacy shares present when appropriate
SMB sessions	Get-SmbSession	Expected sessions; useful before/after cutover
Open SMB files	Get-SmbOpenFile	No unresolved write locks before final sync where applicable
UNC path check	Test-Path \\alpha3\<share>\<path>	True after production identity activation
DNS check	Resolve-DnsName alpha3	Before cutover: legacy target only. After activation: GDPS target only.
Controlled group move	Move-ClusterGroup -Name <role> -Node <peer>	Use only when authorized for TEST or production validation
Windows service check	Get-Service <service-name>	Expected GDPS service Running/Stopped for phase

IDENTITY ADMINISTRATION: Exact DNS, AD computer-object, SPN/Kerberos, clustered client-access point, and SMB identity creation commands must be documented by the Network/AD/Windows owners before production execution. Do not improvise these values during the cutover window.

Appendix B — Operator Sign-Off Sheet

Gate	Owner	Verifier	Date/Time	Decision
G0 TEST rehearsal complete	_____	_____	____/____/____ ____:____	GO / NO-GO
G1 PROD infrastructure ready	_____	_____	____/____/____ ____:____	GO / NO-GO
G2 Production freeze achieved	_____	_____	____/____/____ ____:____	GO / NO-GO
G3 Final synchronization validated	_____	_____	____/____/____ ____:____	GO / NO-GO
G4 Legacy identities released	_____	_____	____/____/____ ____:____	GO / NO-GO
G5 GDPS identities activated	_____	_____	____/____/____ ____:____	GO / NO-GO
G6 Consumer validation passed	_____	_____	____/____/____ ____:____	GO / NO-GO
G7 Production released	_____	_____	____/____/____ ____:____	GO / NO-GO